📘 AI/ML Internship – Day 62 Notes & Tasks

🟢 Module 8: Building an NLP-Based Chatbot using NLTK

📌 Part 1: Recap of Day 61

Yesterday we learned:

✅ Natural Language Processing (NLP)

✅ Intent Recognition

✅ Entity Recognition

✅ Tokenization

✅ Stop Word Removal

✅ Stemming

✅ Lemmatization

Today we will build our **first NLP-based chatbot** using the **NLTK (Natural Language Toolkit)** library.

📌 Part 2: What is NLTK?

Definition

NLTK (Natural Language Toolkit) is one of the most popular Python libraries for Natural Language Processing.

It provides tools to process and analyze human language.

Simple Definition

NLTK helps Python understand text and language.

Features of NLTK

✅ Tokenization

✅ Stop Word Removal

✅ Stemming

✅ Lemmatization

✅ Part-of-Speech (POS) Tagging

✅ Named Entity Recognition (NER)

✅ Text Classification

📌 Part 3: Why Use NLTK in Chatbots?

Suppose a user types:

Can you help me book a train ticket?

Without NLP:

The chatbot sees only text.

With NLTK:

The chatbot can:

- Split the sentence
- Remove unnecessary words
- Find important keywords
- Understand the user's intent

📌 Part 4: Installing NLTK

Install using pip:

```
pip install nltk
```

Import NLTK:

```
import nltk
```

Download required resources:

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

📌 Part 5: Tokenization using NLTK

Definition

Tokenization splits a sentence into individual words.

Example:

Sentence:

I love Artificial Intelligence.

Python Code:

```
from nltk.tokenize import word_tokenize

text = "I love Artificial Intelligence"

tokens = word_tokenize(text)

print(tokens)
```

Output:

['I', 'love', 'Artificial', 'Intelligence']

📌 Part 6: Stop Word Removal

Import stop words:

```
from nltk.corpus import stopwords
```

Example:

```
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords

text = "I want to book a train ticket"

tokens = word_tokenize(text)

stop_words = set(stopwords.words("english"))

filtered = []
```

```
for word in tokens:
    if word.lower() not in stop_words:
        filtered.append(word)
print(filtered)
```

Output:

['want', 'book', 'train', 'ticket']

📌 **Part 7: Stemming using NLTK**

```
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
print(stemmer.stem("playing"))
print(stemmer.stem("running"))
```

Output:

play

run

📌 **Part 8: Lemmatization using NLTK**

```
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
print(lemmatizer.lemmatize("running"))
print(lemmatizer.lemmatize("children"))
```

Output:

running

child

Note: To convert "running" to "run", specify the part of speech:

lemmatizer.lemmatize("running", pos="v")

Output:

run

📌 **Part 9: Building a Simple NLP Chatbot**

```
from nltk.tokenize import word_tokenize

print("🤖 AI Chatbot")
print("Type 'bye' to exit.")

while True:

    user = input("You: ").lower()

    if user == "bye":

        print("Bot: Goodbye!")

        break

    tokens = word_tokenize(user)

    if "hello" in tokens or "hi" in tokens:

        print("Bot: Hello! How can I help you?")

    elif "fee" in tokens or "fees" in tokens:

        print("Bot: The course fee is ₹15,000.")

    elif "duration" in tokens:

        print("Bot: Course duration is 3 months.")

    elif "contact" in tokens:

        print("Bot: Contact us at 9876543210.")
```

```
    else:
        print("Bot: Sorry! I couldn't understand.")
```

📌 Part 10: Why This is Better Than Day 59 Chatbot

Old Chatbot:

Only accepts:

fees

If user types:

Can you tell me the fees?

❌ May fail.

NLP Chatbot:

Splits sentence into words.

Finds:

fees

Replies correctly.

📌 Part 11: Chatbot Workflow

User Message

↓

Lowercase

↓

Tokenization

↓

Stop Word Removal

↓

Keyword Matching

↓

Generate Response

↓

Reply to User

📌 Part 12: Real-Life Example

User:

Can I know the course duration?

↓

Lowercase

↓

Tokenization

↓

Remove Stop Words

↓

Important Words:

course

duration

↓

Chatbot understands:

Intent:

Course Duration

↓

Response:

Course duration is 3 months.

📌 **Part 13: Real-World Applications**

Customer Support

Answer FAQs.

Banking

Balance enquiry

Loan details

Hospital

Appointment booking

Doctor availability

College Admission

Course details

Fee information

Admission process

E-commerce

Order tracking

Return requests

📌 **Part 14: Advantages of NLTK**

✅ Easy to learn

✅ Open source

✅ Large community

✅ Rich NLP tools

✅ Excellent for beginners

📌 Part 15: Limitations

❌ Rule-based understanding

❌ Not as powerful as modern transformer models

❌ Limited contextual understanding

❌ Requires manual keyword creation

📌 Part 16: Difference Between Rule-Based Chatbot and NLP Chatbot

Rule-Based Chatbot	NLP Chatbot
Exact sentence matching	Keyword and language processing
Limited flexibility	More flexible
No text preprocessing	Uses NLP preprocessing
Easy to build	Slightly more advanced
Less accurate	More accurate

📌 Part 17: Theory Questions

What is NLTK?

NLTK is a Python library used for Natural Language Processing tasks.

Why is NLTK used in chatbots?

To process user text using tokenization, stop word removal, stemming, and lemmatization.

What is Tokenization?

Breaking text into individual words or tokens.

Why remove stop words?

To remove common words that add little meaning and focus on important keywords.

What is the advantage of an NLP chatbot over a rule-based chatbot?

An NLP chatbot can understand user input more flexibly instead of relying only on exact matches.

📌 Part 18: Interview Questions

What is NLTK?

A Python library for Natural Language Processing.

Name four NLP techniques available in NLTK.

- Tokenization
- Stop Word Removal
- Stemming
- Lemmatization

Why is tokenization important?

It breaks text into smaller units that are easier for computers to analyze.

What is the difference between stemming and lemmatization?

Stemming removes suffixes, while lemmatization converts words to their proper dictionary form.

How does an NLP chatbot improve user interaction?

By understanding keywords and processing natural language instead of requiring exact phrases.

🎯 Tasks for Day 62

🧠 Theory

1. Define NLTK.
2. Explain Tokenization with an example.
3. Explain Stop Word Removal.
4. Differentiate Stemming and Lemmatization.
5. Compare Rule-Based and NLP-Based Chatbots.

💻 Practical Task 1

Install the NLTK library.

💻 Practical Task 2

Write a Python program to tokenize the sentence:

Artificial Intelligence is changing the world.

💻 Practical Task 3

Remove stop words from:

I want to book a train ticket to Delhi.

💻 Practical Task 4

Apply stemming to:

- Playing
- Running
- Reading
- Learning

💻 **Practical Task 5**

Build a chatbot that answers questions about:

- Fees
- Course Duration
- Eligibility
- Contact Number
- College Location

using NLTK tokenization.